

Notes of HDP¹

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Outline

Introduction

Concentration inequalities

Variance Bounds

- Tensorization property of the Variance
- Bounded Difference Inequalities, Variance
- Poincaré Inequalities

Subgaussian Concentration

- Subgaussianity
- Bounded Difference Inequalities, Subgaussian weak version
- Tensorization property of the Entropy
- Bounded Difference Inequalities, Subgaussian strong version
- Log-Sobolev Inequalities

Sub-Exponential Concentration

Suprema and Chaining

- Finite Maxima
- The Covering Method
- The Chaining Method

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Intro

Informal version of

- ▶ Concentration
- ▶ Suprema

Preliminaries and Notations

- ▶ $a \lesssim b$: $a \leq Cb$ where C is some absolute constant.

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Intro - Concentration inequalities

(1) Variance version of concentration

$$\text{Var}(f) \lesssim \sum_{i=1}^n \mathbb{E} [(D_i f)^2]$$

the variance of a function f of independent (or weakly dependent) random variables is small if the “gradient” of f is small.

(2) Sub-Gaussianity, a sharper control on the distribution of the fluctuations

$$\|f - \mathbb{E}f\|_{\psi_2}^2 \lesssim \left\| \sum_{i=1}^n D_i f \right\|_{\infty}^2 \quad \text{by simple martingale}$$

$$\|f - \mathbb{E}f\|_{\psi_2}^2 \lesssim \left\| \sum_{i=1}^n D_i f \right\|_{\infty}^2 \quad \text{by entropy method}$$



(3)

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Tensorization property of the Variance

Efron-Stein

- relationship between the variance of the overall function and the sum of the individual expected variance[] contributions

[Efron-Stein] Let $X_1, \dots, X_n$ be independent random variables, define

$$\text{Var}_i f(x_1, \dots, x_n) := \text{Var}[f(x_1, \dots, x_{i-1}, \mathbf{x}_i, x_{i+1}, \dots, x_n)] = \text{Var}(f \mid X_{-i})$$

Then

$$\text{Var}[f(X_1, \dots, X_n)] \leq \sum_{i=1}^n \mathbb{E} [\text{Var}_i f(X_1, \dots, X_n)]$$

Bounded Difference Inequalities

Variance version

- relationship between variance of overall function and the sum of Difference $D_i f$ with respect to individual variable

[Bounded Difference Inequalities] Let $X_1, \dots, X_n$ be independent random variables, define

$$D_i f(x_1, \dots, x_n) = \sup_{x_i} f - \inf_{x_i} f$$

$$D_i^- f(x_1, \dots, x_n) = f - \inf_{x_i} f$$

Then

$$\text{Var}(f) \leq \frac{1}{4} \sum_{i=1}^n \mathbb{E} [(D_i f)^2]$$

$$\text{Var}(f) \leq \frac{1}{4} \sum_{i=1}^n \mathbb{E} [(D_i^- f)^2]$$

Example:

$D_i f$, a 'discrete derivative' of the function with respect to the i -th variable

Poincaré Inequalities

- ▶ from 'variance/difference' to 'derivative'

$$\text{Var}(f) \lesssim \mathbb{E} [\|\text{gradient}(f)\|^2]$$

[Convex Poincaré Inequalities] Let $\{X_i\}_{i=1}^n$ be independent random variables, $X_i \in [a, b]$, $f : \mathbb{R}^n \rightarrow \mathbb{R}$ convex and differentiable. Then

$$\text{Var}(f) \leq (b - a)^2 \cdot \mathbb{E} \|\nabla f(X_1, \dots, X_n)\|_2^2$$

proof: first relate 'difference' with 'derivative' with respect to certain assumption on X and f , then use efron-stein/bounded difference inequalities

Poincaré Inequalities

General version

Poincaré Inequalities

Application to Sampling

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Subgaussian Concentration

- ▶ Subgaussianity: another way to describe concentration
- ▶ Entropy Method
- ▶ Log-Sobolev Inequalities

Tail probability bounded by 'Moments'

Chebyshev inequality, Chernoff bound

Let X be a random variable. Then, for any $t > 0$, we have

- ▶ -[Chebyshev inequality]

$$\mathbb{P}\{|X - \mathbb{E}X| \geq t\} \leq \frac{\mathbb{E}|X - \mathbb{E}X|^p}{t^p}$$

- ▶ -[Chernoff bound, exponential moment method] Define the log-moment generating function ψ of a random variable X and its Legendre dual ψ^* as

$$\psi(\lambda) := \log \mathbb{E}e^{\lambda(X - \mathbb{E}X)}, \quad \psi^*(t) = \sup_{\lambda \geq 0} \{\lambda t - \psi(\lambda)\}.$$

Then

$$\mathbb{P}\{X - \mathbb{E}X \geq t\} \leq e^{-\psi^*(t)} \quad \text{for all } t \geq 0.$$

bounding tail probability $\rightarrow$ bounding the MGF.

proof: by Markov Inequality on nonnegative increasing function of $X - \mathbb{E}X$

Subgaussian Variable

[Subgaussian properties] Let X be a random variable. The follows are equivalent, with $K_i > 0$ differing by at most an absolute constant.

(i) (Tail bound) There exists $K_1 > 0$ such that

$$\mathbb{P}\{|X| \geq t\} \leq 2 \exp(-t^2/K_1^2) \quad \text{for all } t \geq 0.$$

(ii) (Moment growth bound) There exists $K_2 > 0$ such that

$$\frac{\|X\|_{L^p}}{\sqrt{p}} = \frac{(\mathbb{E}|X|^p)^{1/p}}{\sqrt{p}} \leq K_2 \quad \text{for all } p \geq 1.$$

[Subgaussian norm] $\|X\|_{\psi_2} = \sup_{p \geq 1} \left\{ \frac{(\mathbb{E}|X|^p)^{1/p}}{\sqrt{p}} \right\}$, the smallest K_2 .

(iii) (Finite MGF of X^2) There exists $K_3 > 0$ such that

$$\mathbb{E} \exp(X^2/K_3^2) \leq 2.$$

Moreover, if $\mathbb{E}X = 0$, (i)–(iii) are equivalent to the following one:

(iv) (MGF bound) There exists $K_4 > 0$ such that

$$\mathbb{E} \exp(\lambda X) \leq \exp(\lambda^2 K_4^2) \quad \text{for all } \lambda \in \mathbb{R}.$$

[Centering] Any subgaussian random variable X satisfies

$$\|X - \mathbb{E}X\|_{\psi_2} \leq C \|X\|_{\psi_2}.$$

Subgaussianity

a centered version

[Subgaussianity] A random variable X is σ^2 -subgaussian if

$$\mathbb{E}e^{\lambda(X-\mathbb{E}X)} \leq e^{\lambda^2\sigma^2/2}, \quad \forall \lambda \in \mathbb{R}$$

If X is σ^2 -subgaussian, then

$$\mathbb{P}\{X - \mathbb{E}X \geq t\} \leq e^{-t^2/2\sigma^2}, \quad \mathbb{P}\{X - \mathbb{E}X \leq -t\} \leq e^{-t^2/2\sigma^2}.$$

This is also equivalent to

$$\|X - \mathbb{E}X\|_{\psi_2}^2 \lesssim \sigma^2.$$

σ^2 is called the variance proxy because it shows like variance, the sum of n independent σ^2 -subg variable is $\frac{1}{n}\sigma^2$ -subg.

Subgaussianity

What r.v.'s are subgaussian?

- ▶ Gaussian
- ▶ A bounded r.v. [Hoeffding] If $a \leq X \leq b$ a.s. for constants a & b , then X is $\frac{(b-a)^2}{4}$ -subg.

Subgaussianity of a function

Bounded Difference Inequalities

Subgaussianity is another description of concentration, and now we discuss the subgaussianity of function $f(X_1, \dots, X_n)$.

[McDiarmid] If $\{X_i\}_{i=1}^n$ are indep, $\|D_i f\|_\infty < \infty$ for all i , then $f(X_1, \dots, X_n)$ is subgaussian with variance proxy $\frac{1}{4} \sum_{i=1}^n \|D_i f\|_\infty^2$

$$\mathbb{P}\{f - \mathbb{E}f \geq t\} \leq e^{-2t^2 / \sum_{i=1}^n \|D_i f\|_\infty^2}$$

$$\|f - \mathbb{E}f\|_{\psi_2}^2 \lesssim \sum_{i=1}^n \|D_i f\|_\infty^2.$$

Actually subgaussian property does not tensorize naturally, and the martingale method can only partially address this issue. Next, we explore the entropy method to derive a better bound.

Entropy and Subgaussian

[Entropy] The entropy of a nonnegative random variable Z is

$$\text{Ent}[Z] := \mathbb{E}[Z \log Z] - \mathbb{E}[Z] \log \mathbb{E}[Z].$$

A sufficient condition for subgaussian in terms of entropy:

[Herbst] If $\text{Ent}(e^{\lambda X}) \leq \frac{\lambda^2 \sigma^2}{2} \mathbb{E} e^{\lambda X}$, $\forall \lambda$, then X is σ^2 -subg.

Tensorization property of the Entropy

Let $X_1, \dots, X_n$ be independent random variables, $f \geq 0$, define

$$g_i(x_1, \dots, x_n) = \text{Ent} [f(x_1, \dots, x_{i-1}, X_i, x_{i+1}, \dots, x_n)]$$

then

$$\text{Ent} [f(X_1, \dots, X_n)] \leq \sum_{i=1}^n \mathbb{E} [g_i(X_1, \dots, X_n)]$$

Bounded Difference Inequalities

a stronger version derived by entropy method

If $\{X_i\}_{i=1}^n$ indep. $\|D_i f\|_\infty < \infty$ for all i , then

$$\|f - \mathbb{E}f\|_{\psi_2}^2 \lesssim \left\| \sum_{i=1}^n |D_i f|^2 \right\|_\infty$$

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Intro - Suprema and Chaining

In previous chapters: $f \rightarrow \mathbb{E}f$

This chapter: about the magnitude of the mean $\mathbb{E}f$

Suprema: focus on a class of f

$$f = \sup_{t \in T} X_t$$

discuss the relation of $\mathbb{E}(\sup_{t \in T} X_t)$ and complexity of T .

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Finite Maxima

Suppose $|T| < \infty$

- ▶ $\mathbb{E} \left(\sup_{t \in T} X_t \right) \leq \sum_{t \in T} \mathbb{E} |X_t| \leq |T| \cdot \sup_{t \in T} \mathbb{E} |X_t|$
- ▶ if X_t have p -th bounded moment,
$$\mathbb{E} \left(\sup_{t \in T} X_t \right) \leq \left[\mathbb{E} \left(\sup_{t \in T} |X_t|^p \right) \right]^{\frac{1}{p}} \leq |T|^{\frac{1}{p}} \cdot \sup_{t \in T} (\mathbb{E} |X_t|^p)^{\frac{1}{p}}$$
- ▶ -[Maximal inequality] Suppose that $\log \mathbb{E}[e^{\lambda X_t}] \leq \psi(\lambda)$ for all $\lambda \geq 0$ and $t \in T$, where ψ is convex and $\psi(0) = \psi'(0) = 0$. Then

$$\mathbb{E} \left(\sup_{t \in T} X_t \right) \leq (\psi^*)^{-1} \log |T|$$

- ▶ if X_t is σ^2 -subg, then

$$\mathbb{E} \left(\sup_{t \in T} X_t \right) \leq \sqrt{2\sigma^2 \log |T|}$$

- ▶ -[Maximal tail inequality]

Application of finite maxima

Model Selection Wang 2026

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The Covering Method

Let T_ε be an ε -cover of T , $\forall t \in T$, $\exists \pi_0(t) \in T_\varepsilon$, s.t. $d(t, \pi_0(t)) \leq \varepsilon$

$$\begin{aligned}\mathbb{E} \left(\sup_{t \in T} X_t \right) &\leq \inf_{\varepsilon > 0} \left\{ \mathbb{E} \left(\sup_{t \in T} X_{\pi_0(t)} \right) + \mathbb{E} \left(\sup_{t \in T} |X_t - X_{\pi_0(t)}| \right) \right\} \\ &\leq \inf_{\varepsilon > 0} \left\{ \mathbb{E} \left(\sup_{t \in T} X_{\pi_0(t)} \right) + \varepsilon \mathbb{E} C \right\} \quad (\text{C-Lip process}) \\ &\leq \inf_{\varepsilon > 0} \left\{ (\psi^*)^{-1} \log |T_\varepsilon| + \varepsilon \mathbb{E} C \right\} \quad (\text{finite maxima})\end{aligned}$$

How to bound $\mathbb{E} \left(\sup_{t \in T} X_t \right)$ if $|T|$ is infinite? **Discretization.**

- ▶ We need the continuity of the random process $\{X_t\}_{t \in T}$. [Lipschitz]
- ▶ And find finite representatives of X_t . How large is the finite representatives? [covering number]
- ▶ so that for arbitrary $t \in T$ we can represent $\sup$
- ▶ apply the finite maximal ineq. to an optimal ε -net.

Covering Number

[Covering number] Consider a metric space (T, d)

- A finite set $T_\varepsilon \subset T$ is called an ε -**net** (also called ε -cover) of T if (every point in T_ε is within distance ε of some point of T)

$$\forall t \in T, \exists \pi(t) \in T_\varepsilon \text{ s.t. } d(t, \pi(t)) \leq \varepsilon.$$

- The smallest cardinality of an ε -net for (T, d) is called the **covering number**. (a measure of the complexity of the set T at the scale ε)

$$N(T, d, \varepsilon) := \inf\{|T_\varepsilon|\}.$$

[Metric entropy] $\log N(T, d, \varepsilon)$

[Covering number of unit ball] $B_d(x, r) := \{y \in \mathbb{R}^d : \|y - x\|_2 \leq r\}.$

Then,

$$\left(\frac{1}{\varepsilon}\right)^d \leq N(B_d(0, 1), \|\cdot\|_2, \varepsilon) \leq \left(1 + \frac{2}{\varepsilon}\right)^d, \quad \forall \varepsilon > 0.$$

Continuity of a random process

Lipschitz process

[Lipschitz process] The random process $\{X_t\}_{t \in T}$ is called C -Lipschitz for a metric d on T if there exists a random variable C such that

$$|X_t - X_s| \leq Cd(t, s) \quad \text{for all } t, s \in T.$$

Example:

- ▶ Lipschitz models
- ▶ Linear class

Lipschitz Maximal Inequality

[Lipschitz Maximal Inequality]

Suppose that $\{X_t\}_{t \in T}$ is C -Lip and $\log \mathbb{E}[e^{\lambda X_t}] \leq \psi(\lambda)$ for all $\lambda \geq 0$ and $t \in T$, where ψ is convex and $\psi(0) = \psi'(0) = 0$. Then,

$$\mathbb{E} \left(\sup_{t \in T} X_t \right) \leq \inf_{\varepsilon > 0} \left\{ (\psi^*)^{-1} [\log N(T, d, \varepsilon)] + \varepsilon \cdot \mathbb{E} C \right\}.$$

► if X_t is σ^2 -subg, then

$$\mathbb{E} \left(\sup_{t \in T} X_t \right) \leq \inf_{\varepsilon > 0} \left\{ \sqrt{2\sigma^2 [\log N(T, d, \varepsilon)]} + \varepsilon \cdot \mathbb{E} C \right\}.$$

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The Chaining Method

the one-resolution discretization may give us sub-optimal bounds of the generalization gap.

How to bound $\mathbb{E} \left(\sup_{t \in \mathcal{T}} X_t \right)$ if $|\mathcal{T}|$ is infinite? And the random process $\{X_t\}_{t \in \mathcal{T}}$ is Lipschitz "in probability"?

Continuity of a random process

ψ process, separable process

Dudley

multi-resolution approximation

[Dudley] Let $\{X_t\}_{t \in T}$ be a separable Ψ -proc. on a metric space (T, d) .
Then,

$$\mathbb{E} \left(\sup_{t \in T} X_t \right) \leq 3 \times \sum_{k \in \mathbb{Z}} 2^{-k} \cdot (\psi^*)^{-1} (2 \log N(T, d, 2^{-k}))$$

(Dudley's integral inequality)

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Proof of Efron-Stein

Proof of Hoeffding

Proof of McDiarmid

references

Wang, Kaizheng (2026). “Pseudo-labeling for kernel ridge regression under covariate shift”. In: *The Annals of Statistics* 54.1, pp. 252–276.